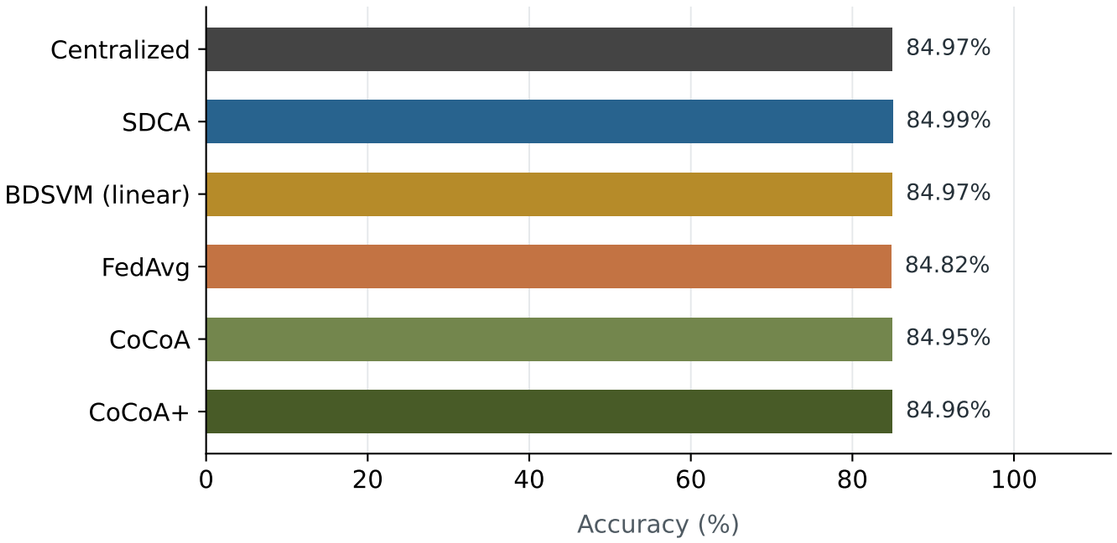
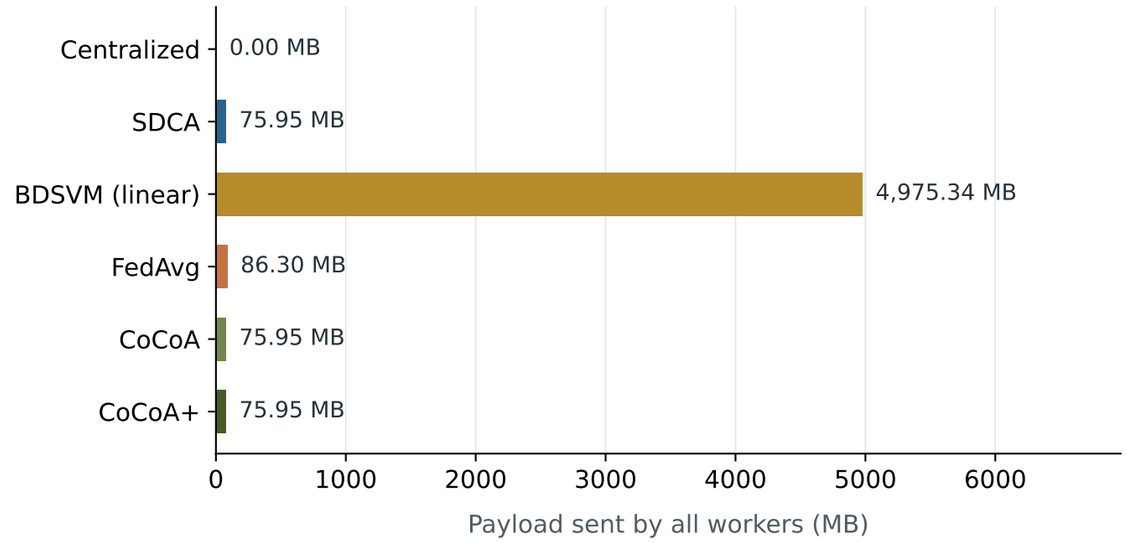
a9a · comparison preview

32,561 train / 16,281 test samples · 10 workers · seed 42 · G=50 rounds · relative tolerance 0.001

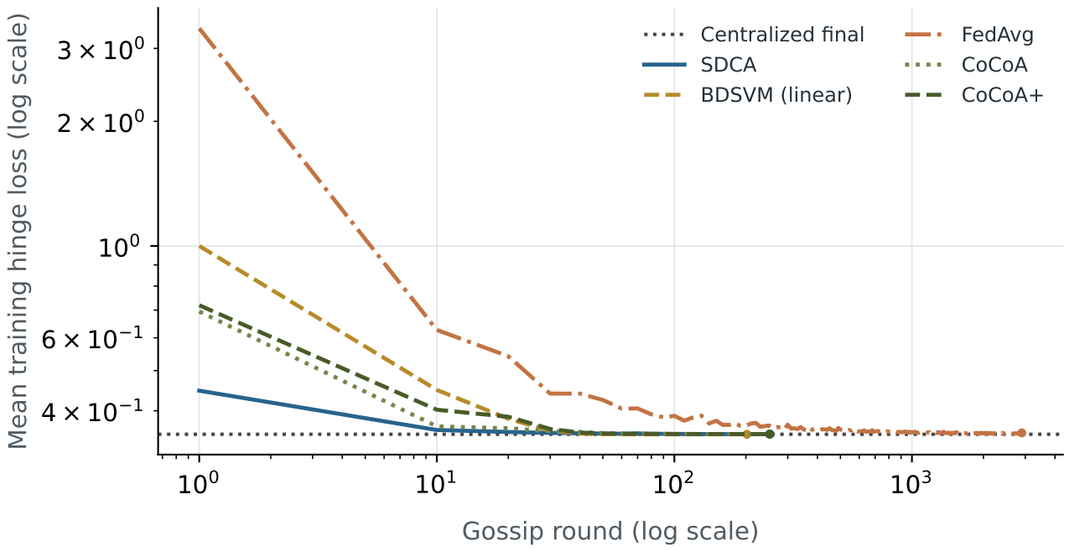
Final test accuracy



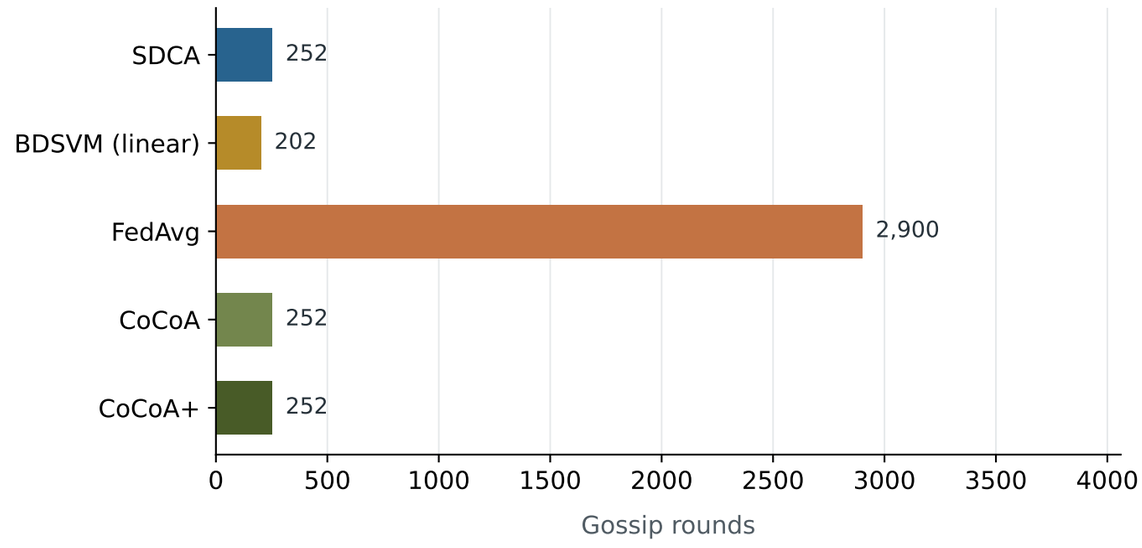
Total communication at training stop



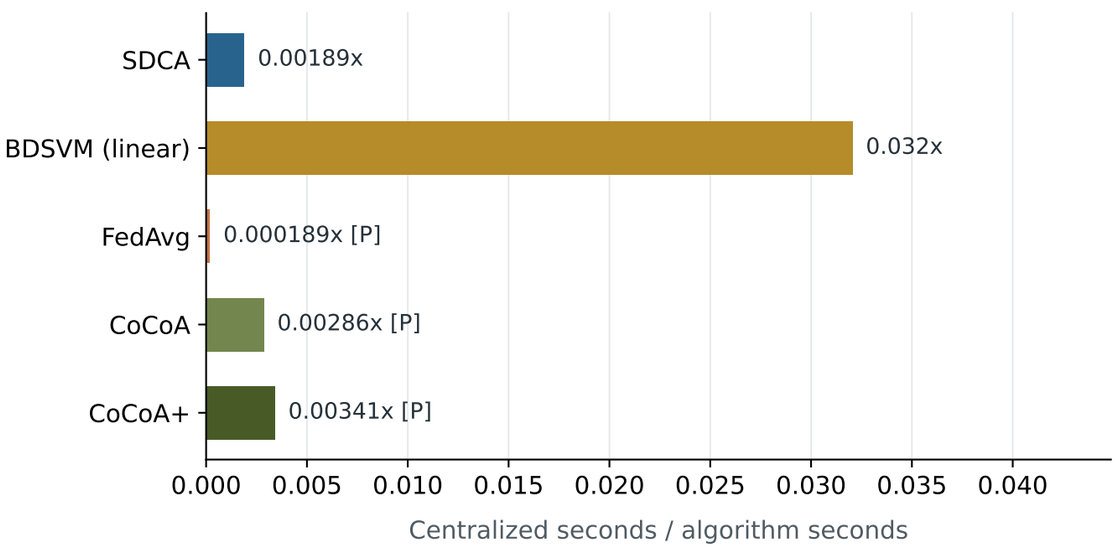
Hinge loss on the training set



Training rounds until stop



Observed time ratio versus centralized baseline



Training time and stopping status

Method	Seconds	Stop
Centralized	0.205	Solver tolerance
SDCA	108.477	Train plateau
BDSVM (linear)	6.411	Train plateau
FedAvg	1,088.293	Train plateau [P]
CoCoA	71.782	Train plateau [P]
CoCoA+	60.265	Train plateau [P]

1x = centralized time; values below 1x are slower.
[P] Ratios use elapsed time from overlapping runs.
Other timings use sequential simulation on one host.
Includes train monitoring; this is not a multi-machine speedup.

Test accuracy is evaluated only after passing the training stop rule. Hatched bars indicate the iteration limit, not convergence.
SSL is excluded from this figure. Time ratios describe observed elapsed time, not a fair isolated speedup comparison.
All P2P methods: adjacent 50-round windows, all peers, two stable comparisons. Communication counts directed numerical payloads, excluding headers and offline monitoring.
Models, objectives and local compute budgets differ. FDR-SVM is not integrated and is omitted. [P] FedAvg, CoCoA and CoCoA+ overlapped in this run.